Mohannad Shafik

Junior Full-Stack Developer — AI & Automation

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PROFILE

Junior full-stack developer from Turnhout focused on AI and automation. I build end-to-end systems in Python and TypeScript: API, database, dashboard and AI integration via the Claude API. My projects are on GitHub; the webhook dashboard has an online demo. I'm looking for a first role as a junior developer where I can contribute quickly and keep growing. No paid experience, but projects with tests, CI and an online demo.

PROJECTS

Webhook platform (ms-webhook-api + ms-webhook-ui + ms-webhook-api-dotnet)
Python, FastAPI, PostgreSQL, Redis, Docker · C#, ASP.NET Core 8, EF Core · React 18, TypeScript, TanStack Query, Tailwind

- Asynchronous webhook delivery with retry logic and HMAC signing
- Real-time dashboard showing deliveries; mock mode for a demo without a backend
- Same API rebuilt in C# / ASP.NET Core 8: same routes and JSON, 70 automated tests, dashboard checked against both versions

Knowledge assistant (RAG) (ms-knowledge-assistant)
Python, FastAPI, PostgreSQL + pgvector, React, Docker

- Answers questions about company documents (PDF, DOCX, Markdown) with cited sources, and says 'I don't know' when the documents don't contain the answer
- Local embeddings; optionally a local model (Ollama) or Claude writes the answer; measured with a 38-question evaluation set

Claude API tools (ms-ai-review + ms-agent)
Python, Claude API

- CLI tool for code review via the Claude API, with streaming and prompt caching
- Agent (ReAct pattern) that analyses a codebase, with tool use and self-validation

Data quality tool (ms-data-quality)
Python, openpyxl

- Finds invalid emails, duplicate rows and company-name variants in CSV/Excel and applies only what the user approves
- Optionally a language model picks the right company name; 63 automated tests

Telegram signal platform (private prototype)
Python, PostgreSQL, Redis, React

- Prototype: distributes signals to multiple Telegram bots with a queue and subscriptions; private repository, not in production, no tests

EXPERIENCE

Personal projects — 2023 – present · Turnhout

- Built and documented six applications, with no paying clients; see Projects

IT technician internship, Level Academy, Birkirkara (Malta) — June 2023 · 2 weeks

- Set up network routing and remote desktop for a gaming room with multiple PCs
- Assembled workstations and installed peripherals
- Installed and tested security cameras

EDUCATION

Higher education, ICT / Programming, not completed (no degree) — started Sep 2023

Computer Management (ICT), Campos Turnhout — Jan 2022 – Jun 2023 · EQF level 4 · diploma obtained

- Software, hardware and office software

SKILLS

Hands-on: Python, TypeScript/JavaScript, React, FastAPI, PostgreSQL (pgvector), Redis, Docker, Git/GitHub, REST APIs, C#/.NET 8 (one project)
AI: Claude API (agent and tool-use code written and tested), prompt design, RAG and embeddings
Also: SQL, HTML/CSS, Tailwind, Nginx, pytest, GitHub Actions
School-level: Java, PHP

LANGUAGES

Dutch (native) · Arabic (native) · English (C2) · French (B1)